MATHS MANTHAN
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Subject: Mathematics **PERIODIC ASSESSMENT TEST – 09 (LINEAR EQUATIONS)** **Time:** 60 Minutes

Class: Class X (CBSE) **Chapter 3: Pair of Linear Equations in Two Variables** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 14(iv).
2. **Sec A:** 4 MCQs (1M). **Sec B:** 2 AR (1M). **Sec C:** 3 VSAQ (2M). **Sec D:** 3 SA (3M). **Sec E:** 1 LA (5M). **Sec F:** 1 Case Study (4M).

SECTION A: Multiple Choice Questions (4 × 1 = 4 Marks)

Q1. The graph of the linear equation $x = -2$ is a line: [1]
(A) parallel to the x-axis (B) parallel to the y-axis (C) passing through origin (D) perpendicular to y-axis

Q2. The graph of the linear equation $y = 5$ is a line: [1]
(A) parallel to the x-axis (B) parallel to the y-axis (C) inclined at $45°$ to axes (D) passing through origin

Q3. The pair of equations $x = a$ and $y = b$ graphically represents lines which are: [1]
(A) parallel (B) intersecting at (b, a) (C) coincident (D) intersecting at (a, b)

Q4. If $2x - 3y = 7$ and $(a + b)x - (a + b - 3)y = 4a + b$ represent coincident lines, then a and b satisfy: [1]
(A) $a + 5b = 0$ (B) $5a + b = 0$ (C) $a - 5b = 0$ (D) $5a - b = 0$

SECTION B: Assertion-Reason Questions (2 × 1 = 2 Marks)

Directions: (A) Both (A) and (R) are true, and (R) is correct explanation. (B) Both true, but (R) is not correct explanation. (C) (A) is true, (R) is false. (D) (A) is false, (R) is true.

Q5. **Assertion (A):** The graph of the equation $2x + 3y = 6$ does not pass through the origin $(0, 0)$.
Reason (R): A straight line $ax + by + c = 0$ passes through the origin if and only if $c = 0$. [1]

Q6. **Assertion (A):** The system of equations $x + y = 8$ and $x - y = 2$ represents intersecting lines.
Reason (R): A pair of linear equations represents intersecting lines if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. [1]

SECTION C: Very Short Answer Questions (3 × 2 = 6 Marks)

Q7. If $x = a$ and $y = b$ is the solution of the equations $x - y = 2$ and $x + y = 4$, find the values of a and b. [2]

Q8. Check whether the lines $x - 2y = 0$ and $3x + 4y - 20 = 0$ intersect at a unique point, and find the point algebraically. [2]

Q9. Find the value of k for which the system of equations $x - ky = 2$ and $3x + 2y = -5$ has a unique solution. [2]

SECTION D: Short Answer Questions (3 × 3 = 9 Marks)

Q10. In a $\triangle ABC$, $\angle C = 3\angle B = 2(\angle A + \angle B)$. Find the measures of all three angles. [3]

Q11. Find the values of a and b for which the following system of linear equations has infinitely many solutions: [3]

$$2x + 3y = 7 \quad \text{and} \quad (a - b)x + (a + b)y = 3a + b - 2$$

Q12. Draw the graphs of $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Determine the coordinates of the vertices of the triangle formed by these lines and the y-axis, and calculate its area. [3]

SECTION E: Long Answer / HOTS Question (1 × 5 = 5 Marks)

Q13. Draw the graphs of the linear equations $2x - y = 1$ and $x + 2y = 13$. Find the coordinates of the point of intersection of the lines, the coordinates of the vertices of the triangle formed by these lines and the line $y = 0$, and calculate the area of this triangle. [5]

SECTION F: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q14. Case Study: Sports Academy Equipment Procurement

The sports department of a school is preparing for an inter-school tournament. The head coach placed two consignments with an equipment dealer:
First consignment: 7 cricket bats and 6 leather balls for Rs. 3800.
Second consignment: 3 cricket bats and 5 leather balls of the same quality for Rs. 1750.
Let the price of one bat be Rs. x and the price of one ball be Rs. y.

Based on the above procurement details, answer the following questions:

(i) Represent the two purchases algebraically as a pair of linear equations in x and y. [1]

(ii) Find the unit price of a single cricket bat (x). [1]

(iii) Find the unit price of a single cricket ball (y). [1]

(iv) What would be the total bill if the coach places an order for 5 bats and 4 balls? [1]

OR

Find the difference between the price of 2 bats and 10 balls. [1]

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